

Smart Scan Strategy for Electronic Warfare: A Comprehensive Analysis and Implementation Framework

The Electromagnetic Battlespace and the Strategic Imperative for Autonomous Scheduling

The contemporary electromagnetic spectrum represents a congested, contested, and highly dynamic operational domain. Within this environment, Electronic Support Measures and Radar Warning Receivers are tasked with the critical function of detecting, intercepting, and analyzing hostile communication and radar signals before adversaries can establish a tactical advantage. Historically, these systems have relied on open-loop scanning strategies. These traditional paradigms utilize pre-programmed, deterministic sweep patterns based on prior electronic intelligence and pre-mission data to cycle through frequency bands. While an open-loop strategy provides a mathematical guarantee that the entire operational frequency band is swept over a given time interval, it is fundamentally inefficient in dynamic threat environments. The open-loop methodology allocates equal temporal resources to empty frequency bands, benign civilian transmissions, or non-threatening emitters, thereby losing critical observation time that should be dedicated to tracking agile, highly lethal, or newly emerging threats. The problem statement SIH26055, articulated by the Defence Research and Development Organisation under the iDEX initiative, addresses this specific vulnerability in legacy Electronic Warfare architectures¹. The mandate requires the development of a "Smart Scan Strategy for Electronic Warfare" that functions optimally even in the absolute absence of prior reliable intelligence regarding hostile emitters and their operating characteristics. This requirement fundamentally transitions the interception task from a static scheduling process to a dynamic, two-dimensional search problem spanning both frequency and time¹. The primary objective is the development of robust, machine learning-driven receiver scheduler software that operates as a closed-loop system. This software must dynamically learn the local radio frequency environment, predict the behavior of spatially scanning and frequency-agile emitters, and autonomously optimize its dwell times and revisit rates.

By treating signal interception as a complex optimization problem constrained by limited receiver resources, the software must maximize defined figures of merit. These include the probability of detection, the average intercept rate, and the percentage of correct predictions, while simultaneously minimizing the average intercept time error and the probability of false alarms¹. The realization of this system demands a synthesis of advanced digital signal processing, reinforcement learning, and stochastic modeling to construct an autonomous cognitive receiver framework capable of outmaneuvering modern, software-defined radar systems.

Physics of Interception and Electronic Support Receiver Architecture

To construct an artificial intelligence-driven scanning scheduler, it is first necessary to mathematically define the physical constraints of Electronic Support receivers and the mechanics of signal interception. The Probability of Intercept is the foundational metric of any such system. It dictates the likelihood that an adversary's transmitted signal will coincide spatially, temporally, and spectrally with the receiver's observation window⁵.

Because broadband wide-open receivers, such as Crystal Video Receivers, suffer from exceptionally low sensitivity and high noise floors, modern systems overwhelmingly utilize narrowband Scanning Superheterodyne receivers⁶. These receivers sweep a narrow instantaneous bandwidth across a wide total frequency range. While this architecture dramatically increases sensitivity and allows for the detection of faint signals at extended ranges, it concurrently reduces the temporal window during which any specific frequency band is observed⁷.

An interception event in this architecture requires the simultaneous alignment of three distinct window functions. First, frequency coincidence must occur, requiring the receiver's tuned frequency band to encompass the emitter's instantaneous transmission frequency⁴. Second, spatial coincidence is required, meaning the emitter's directional main beam or significant side lobes must illuminate the physical location of the receiver¹⁰. Finally, temporal coincidence is mandatory; the receiver must be actively dwelling on the correct frequency precisely when the radar pulse arrives¹⁰.

Mathematically, a deterministic interception during a single period is governed by the overlap of these activity windows¹⁰. The deterministic parameters involve the target emitter and the scanning receiver dynamics. Assuming the starting times of the emitter's pulse train and the receiver's scanning cycle are statistically independent, the cumulative probability of intercept within a specified observation time can be modeled mathematically. The scheduler's decision-making process is subsequently reliant on the quality of the intercepted signal, measured by the Signal-to-Noise Ratio³.

Interception Parameter	Mathematical Notation	Operational Definition
Emitter Illumination Time	τ_e	The duration a target radar's main beam sweeps across the receiver.
Emitter Rotation Period	T_e	The time required for the target antenna to complete one full mechanical or electronic sweep.

Receiver Dwell Time	τ_r	The temporal duration the receiver listens to a specific frequency band before tuning elsewhere.
Receiver Sweep Period	T_r	The total time required for the receiver to scan its entire designated frequency spectrum.
Probability of Intercept	$P_{int}(T)$	The probability that an intercept occurs within a total observation window T .

The received power P_r at the receiver antenna is governed by the one-way radar range equation. The ratio of this signal power to the inherent thermal noise of the receiver dictates the Signal-to-Noise Ratio, which is fundamental not merely for initial detection, but as a time-series feature utilized by the artificial intelligence to estimate the rotational period and spatial characteristics of the hostile emitter's antenna pattern³.

$$SNR = \frac{P_t G_t G_r \lambda^2}{(4\pi R)^2 k T_{eff} B F L}$$

Where P_t represents the peak transmitter power, G_t is the emitter antenna gain, G_r is the receiver antenna gain, λ is the operational wavelength, R is the line-of-sight distance to the radar, k is Boltzmann's constant, T_{eff} is the effective noise temperature, B is the receiver bandwidth, F is the noise figure, and L represents combined system and propagation losses¹². An intelligent scanning strategy continuously ingests variations in this Signal-to-Noise Ratio to map the electromagnetic battlespace.

The Threat Landscape: Cognitive and Multi-Function Radars

The strategic necessity for a machine learning-based scheduler is strictly driven by the rapid evolution of adversary radar systems. Traditional scanning strategies were optimized to counter mechanically steered search radars characterized by fixed Pulse Repetition Intervals,

fixed pulse widths, and predictable, continuous mechanical rotation¹⁴. However, the modern threat environment is dominated by Cognitive Radars and Active Electronically Scanned Array-based Multi-Function Radars¹⁶.

Multi-Function Radars are designed to interleave multiple distinct tasks—such as velocity search, range while search, track and search, and single target track—virtually simultaneously by electronically steering their beams in microseconds without the inertia of mechanical gimbals¹⁵. This rapid multiplexing creates an emission profile that completely defeats open-loop scanners.

These modern systems employ extreme Pulse Repetition Interval agility, utilizing staggered, jittered, sliding, and dwell-and-switch modulations¹⁸. This deliberate randomization breaks the periodicity that traditional autocorrelation algorithms rely upon for signal sorting. Furthermore, these radars utilize pulse-to-pulse or burst-to-burst frequency hopping over wide bandwidths to mitigate jamming and evade narrowband interception¹⁹. The integration of Low Probability of Intercept techniques, including the use of ultra-wideband waveforms, continuous wave phase modulation, and strict emission control protocols, further reduces peak power⁶. This intentional blending of the radar signal into the thermal noise floor ensures that receivers utilizing legacy threshold detection fail to register a threat.

To effectively counter this sophisticated threat matrix, the proposed framework relies on an artificial intelligence engine that does not merely react to isolated pulses. Instead, the system models the underlying syntactic language and behavioral state transitions of the Multi-Function Radar to predict its next emission in both the temporal and spectral domains¹.

Simulation Environments and Foundational Datasets

The development and rigorous training of an artificial intelligence-based scheduler necessitate a highly realistic, simulated radio frequency environment. The system models require a simulated ground truth regarding the status of all emitters across all frequency bands and time slots to accurately calculate loss functions and reward updates during training¹. To achieve this, the architecture fuses data from highly specific, open-source intelligence databases and synthetic benchmarks.

The Turing Synthetic Radar Dataset

To train the machine learning models on the complex task of processing raw interceptions, the framework leverages the Turing Synthetic Radar Dataset, engineered by the Alan Turing Institute¹. This dataset serves as the premier benchmark for radar pulse deinterleaving and electronic support research, simulating the extreme density of modern electronic warfare scenarios²².

The dataset contains six thousand discrete pulse trains spanning two receiver configurations—scan mode and stare mode—totaling over four billion individual pulses. It perfectly replicates highly congested scenarios featuring up to one hundred and ten simultaneous emitters with significant parameter space overlap²². Crucially for the scheduler pipeline, the simulated receiver outputs sequences of Pulse Descriptor Words rather than raw

in-phase and quadrature data. Each Pulse Descriptor Word represents a discrete detected pulse and encapsulates multidimensional features: Time of Arrival, Radio Frequency, Pulse Width, Angle of Arrival, and Amplitude¹⁵. The physical realism of the dataset guarantees that models trained upon it understand physical signal propagation effects, such as free-space path loss and antenna beam dynamics, thereby preventing the artificial intelligence from learning impossible physics²².

Parametrization via the JC Wise Radar Emitter Database

While the Turing dataset provides the synthetic pulse structure, the mathematical boundaries of the simulation are constrained by the JC Wise Radar Emitter Database¹. This comprehensive repository catalogues the precise technical specifications, operating bands, scanning mechanics, and deployment doctrines of contemporary military radar systems. By analyzing the data within the JC Wise database—which includes deep technical parameters of systems such as the Russian Kvant SPN-series high-power jammers, the Nebo SVU VHF-band 3D surveillance radars, and advanced Chinese People's Liberation Army phased arrays—the simulation bounds are locked to reality²⁴. The fusion of the static constraints derived from the JC Wise database with the dynamic pulse generation of the Turing dataset creates a robust, heterogeneous, and tactically accurate training ground for the scheduling agent.

Algorithmic Architecture: The Tripartite Smart Scan Strategy

The interception of transient signals in a dynamic, unknown environment represents a sequential decision-making problem under severe uncertainty. The receiver controls a strictly limited resource—observation time—and must allocate it across multiple discrete frequency channels to maximize cumulative intercepts while simultaneously minimizing the cost of missed lethal threats. This operational challenge is modeled mathematically as a Partially Observable Markov Decision Process²⁸.

Because the receiver architecture is inherently narrowband, it is physically impossible to observe the entire electromagnetic spectrum simultaneously. Consequently, the true state of the global environment remains permanently hidden from the agent⁸. The proposed architecture solves this profound lack of visibility through a multi-stage machine learning pipeline. This pipeline comprises Quantum-Inspired Multi-Armed Bandits for initial spectrum exploration, Predictive State Representations for decoding radar behavior, and Dueling Double Deep Q-Networks for executing optimal scheduling policies.

Phase 1: Rapid Spectrum Exploration via Multi-Armed Bandits

At system initialization, the operational mandate dictates an absence of prior reliable intelligence¹. The receiver possesses absolutely no knowledge regarding which frequency bands currently contain active, hostile emitters³. Initiating a purely random or linear sequential scan under these conditions guarantees a failure to detect fast-acting threats. To resolve this,

the spectrum exploration phase is mathematically formulated as a Multi-Armed Bandit problem. In this construct, each discrete frequency sub-band represents an independent "slot machine arm" yielding an unknown payoff probability characterized by the presence of a radar signal³¹.

To balance the classic reinforcement learning trade-off between exploring unvisited frequencies and exploiting bands where signals have already been intercepted, the scheduler employs the Upper Confidence Bound algorithm³². The Upper Confidence Bound mathematically guarantees that the scheduler will rapidly converge on active frequency bands while still periodically checking dormant bands to ensure newly illuminated threats are not

missed³¹. The algorithm selects the next frequency band A_t to scan at time step t according to the maximization of the confidence bound:

$$A_t = \arg \max_a \left(Q_t(a) + c \sqrt{\frac{\ln t}{N_t(a)}} \right)$$

Where $Q_t(a)$ is the empirically estimated value of band a , representing the average Signal-to-Noise Ratio or the historical intercept rate, $N_t(a)$ is the total number of times band a has been visited by the receiver, and c is a tunable hyperparameter strictly controlling the degree of exploration³⁴. As a band is repeatedly sampled, the uncertainty term shrinks, forcing the algorithm to explore other unvisited bands unless the known band's value $Q_t(a)$ is exceptionally high.

Phase 2: Radar Behavioral Modeling using Predictive State Representations

Once active bands are identified and intercepted pulses are collected, the system faces the complex task of decoding the internal state of the hostile Multi-Function Radars to predict their future emission schedules. Traditional stochastic models utilize Hidden Markov Models for this purpose; however, Hidden Markov Models assume an unobservable, abstract hidden state that requires extensive prior intelligence to define and initialize accurately, rendering them brittle against non-cooperative, novel radars¹⁷.

To maintain a truly autonomous, closed-loop architecture, the system utilizes Predictive State Representations to model the adversary¹⁷. Predictive State Representation is a data-driven dynamical system paradigm where the "state" of the system is defined entirely by a set of mathematical predictions about future observable events—specifically, the properties of future Pulse Descriptor Words—conditional exclusively on past observations¹⁷.

The hostile radar's signal sequence is treated as an uncontrolled stochastic process. The

predictive model is defined formally by the tuple $\langle \mathcal{O}, \mathcal{H}, \mathcal{E}, Q \rangle$ ¹⁷. By continuously updating the state vector based solely on real-time intercepted pulse data, the module allows the

scheduler to accurately recognize the radar's operating modes, such as a shift from broad volume search to concentrated target tracking, without relying on pre-mission threat databases¹.

PSR Component	Mathematical Notation	Application in Electronic Support
Observation Space	$\mathcal{O}$	The finite discrete set of all possible extracted PDW features (frequency, pulse width, amplitude variations).
History	$\mathcal{H}$	The sequence of observations intercepted from the initial time to the current time, $h = o_1 o_2 \dots o_t$.
Future Events	$\mathcal{E}$	The anticipated sequence of observations following the history, $e = o_{t+1} o_{t+2} \dots$
Core Events	$\mathcal{Q}$	A minimal subset of future events whose probabilities form a basis capable of expressing the probability of any arbitrary future event through linear combination.

Phase 3: Closed-Loop Scheduling via Dueling Double Deep Q-Networks

With the electromagnetic environment mapped by the Multi-Armed Bandit and the emitter behaviors temporally modeled by the Predictive State Representations, the core operational decisions—which specific frequency to visit next and precisely how long to dwell there—are optimized using Deep Reinforcement Learning³⁷. Given the high-dimensional nature of the environment and the hybrid discrete-continuous nature of frequency selection and dwell time

allocation, the architecture deploys a Dueling Double Deep Q-Network⁴⁰.

The integration of the scheduling problem into the Markov Decision Process requires a highly engineered state, action, and reward formulation to ensure convergence. The State Space (S_t) is a multidimensional vector encoding the current situational awareness. It ingests the Predictive State Representation output vector, the elapsed time since each frequency band was last visited, the history of recent intercepts, and the current receiver hardware tuning latency⁴³. The Action Space (A_t) is formulated as a tuple dictating the next center frequency to tune the receiver to, and the optimal dwell time to remain on that band⁴.

The optimization of the Reward Function (R_t) is the paramount factor ensuring the artificial intelligence adheres to the objectives mandated by the SIH26055 problem statement¹. The multi-objective reward is structured to incentivize positive intercepts while aggressively penalizing inefficiency:

$$R_t = \alpha(H_t) - \beta(M_t) - \gamma(\Delta t_{err}) - \delta(C_{tune})$$

In this formulation, H_t represents successful interceptions weighted proportionally to the threat priority of the emitter, M_t represents the penalty for missing a predicted pulse, Δt_{err} penalizes the average intercept time error when the receiver arrives at a band too early or too late, and C_{tune} represents the hardware latency penalty associated with constantly tuning the local oscillator to disjointed frequencies¹.

Standard Q-learning algorithms suffer from a documented overestimation bias, where the neural network systematically overestimates the value of suboptimal actions due to the max operator in the Bellman equation. The Double Deep Q-Network architecture mitigates this fatal flaw by strictly decoupling action selection from action evaluation using a target network⁴⁸. The Dueling architecture further optimizes the learning rate by splitting the neural network's final layers into two separate, concurrent streams. One stream estimates the State Value

function $V(s)$, quantifying how inherently valuable or dangerous the current overall radio frequency environment is regardless of the receiver's next action. The second stream

estimates the Advantage function $A(s, a)$, quantifying the relative benefit of tuning to a specific frequency band versus all other available actions⁵¹.

The final Q-value for the scheduling decision is aggregated using the following formulation:

$$Q(s, a; \theta, \alpha, \beta) = V(s; \theta, \alpha) + \left(A(s, a; \theta, \beta) - \frac{1}{|\mathcal{A}|} \sum_{a'} A(s, a'; \theta, \beta) \right)$$

Where θ represents the shared convolutional feature extraction layers processing the

time-series data, and α and β are the parameters of the fully connected value and advantage streams, respectively⁵¹. This dueling topology allows the agent to rapidly learn which frequency bands are entirely empty or irrelevant without having to calculate the exact, granular Q-value for every possible dwell time within those empty bands. This architectural efficiency drastically accelerates the convergence of the scheduler during the highly chaotic initial phases of an engagement⁴⁰.

Interception Algorithms for Periodic Scan Receivers

The SIH26055 problem statement explicitly requests the outlining of approaches and the development of algorithms to optimally intercept a periodic scan receiver¹. While modern Multi-Function Radars utilize electronic beam steering, the vast majority of legacy early warning, coastal surveillance, and target acquisition radars globally still rely on periodic mechanical rotation⁴.

For these highly structured, periodic emitters, the artificial intelligence scheduler bypasses the heavy neural network inference and instead utilizes a rapid Signal-to-Noise Ratio Autocorrelation algorithm³. When the Electronic Support receiver dwells on a band and successfully detects pulses, it records the continuous, normalized time-series of the Signal-to-Noise Ratio. As the enemy radar's main beam mechanically sweeps across the physical location of the receiver, the recorded amplitude peaks dramatically. As the antenna sweeps away, the amplitude drops back into the radar's side-lobes and eventually into the noise floor.

The autocorrelation of this intercepted time series, denoted as $\hat{\rho}$, at lag k is calculated computationally to extract the underlying periodicity:

$$\hat{\rho}(k) = \frac{1}{N} \sum_{i=1}^{N-k} (SNR_i - \mu)(SNR_{i+k} - \mu)$$

The temporal distance between the periodic peaks of the resulting autocorrelation function yields the precise mechanical rotation period of the enemy radar's antenna³. Once the scheduling agent mathematically isolates this period, it dynamically injects a hard, deterministic constraint into the action space. It schedules a highly specific, short-duration tracking dwell on that specific frequency band exactly when the peak of the antenna pattern is mathematically predicted to return to the receiver's azimuth. This highly deterministic approach ensures a guaranteed intercept on subsequent rotations with zero wasted dwell time, overcoming the "lock-in step" problem where a receiver's sweep period and a radar's rotation period fall into a malicious synchronization that prevents interception³.

Deinterleaving and Data Preprocessing Pipelines

In a dense, real-world electronic warfare environment, the receiver intercepts millions of individual pulses per second originating from multiple, distinct, and simultaneous radars. This

data stream is heavily interleaved, presenting as a chaotic sequence of pulses. Before the predictive state models or the scheduling networks can analyze a specific target radar, the raw, composite pulse stream must be separated into individual, single-emitter pulse trains. This mandatory preprocessing stage is known as pulse deinterleaving²¹.

The software utilizes a deep learning-based spatial clustering algorithm to accomplish deinterleaving²². Using the Turing Synthetic Radar Dataset as the foundational baseline, a representation learning model maps the raw Pulse Descriptor Words into a high-dimensional latent space. Pulses that originate from the identical physical emitter naturally form dense geometric clusters in this latent space, even if the emitter utilizes complex, staggered pulse repetition intervals designed to confuse sequential sorters. Density-based spatial clustering algorithms group these points, successfully isolating individual emitter tracks from the noise. These purified, single-emitter tracks are then passed sequentially to the scheduling agent for behavioral prediction²³.

Performance Evaluation and Critical Figures of Merit

As explicitly required by the operational parameters of SIH26055, the software scheduler must dynamically build up, track, and maximize specific figures of merit representing interception performance¹. The system logs these metrics continuously during the simulation and validation phases to benchmark the artificial intelligence model against the absolute ground truth of the simulated radio frequency environment.

Performance Metric	Operational Definition and System Relevance
Probability of Detection (P_d)	The ratio of hostile radar emissions successfully intercepted and processed by the receiver versus the total number of valid emissions physically illuminating the receiver. The deep learning scheduler drives this ratio to operational superiority over sequential scanning ¹ .
Average Intercept Rate	Measured as the number of unique, critical emitters intercepted per second. By actively minimizing dwell time on empty bands using the multi-armed bandit algorithm, the intercept throughput is drastically maximized ¹ .
Average Intercept Time Error	Calculated as the mean absolute difference between the predicted time of pulse arrival

	and the actual time of interception. A lower error proves the receiver is tuning to the target band precisely as the pulse arrives, eliminating wasted idle time ¹ .
Probability of False Alarm (P_{fa})	In the context of resource scheduling, a false alarm represents a critical algorithmic failure where the system decides to allocate a long, valuable dwell time to a frequency band incorrectly predicted to contain an active threat ¹ .
Reward Function Convergence	The training telemetry of the neural network continuously tracks the cumulative reward per episode. A successful, battle-ready deployment is indicated by the asymptotic stabilization of this curve, proving the agent has mathematically solved the optimal scheduling policy ¹ .
Percentage of Correct Predictions	The exact ratio of instances where the Predictive State Representation model accurately forecasts the next operating mode transition of an adversary radar, verifying the system's cognitive tracking capability ¹ .

Deployment Architecture and Edge Computing Execution

While the primary deliverable for the problem statement is a highly advanced software model, the architectural design must rigorously account for the severe physical constraints of tactical deployment on airborne, naval, or terrestrial Electronic Warfare platforms¹.

Electronic warfare engagements require microsecond-level reaction times. A scheduler that requires hundreds of milliseconds of compute time to output the next action is tactically obsolete before the command is even executed. The proposed architecture solves this inherent latency through an asymmetric, edge-computing design philosophy.

The computationally intensive training of the networks—involving the ingestion of billions of simulated pulses, loss calculation, and continuous backpropagation—is conducted entirely offline on high-performance, secure GPU clusters⁵⁶. The resulting trained models are subsequently quantized, pruned, and compiled using optimized frameworks designed for edge inference.

The inference engine, stripped of its training overhead, requires negligible computational resources, executing decision loops in sub-milliseconds on ruggedized, Field-Programmable Gate Array (FPGA) or embedded system-on-chip hardware. This edge module interfaces directly with the receiver's local oscillator controller, translating neural network outputs into immediate physical hardware tuning commands⁴⁴. Furthermore, the software is engineered with a modular, vendor-agnostic Application Programming Interface. It ingests standardized pulse data arrays, ensuring that whether the Department of Defence deploys this algorithm on a fighter aircraft's radar warning receiver or a ground-based signals intelligence vehicle, the scheduler integrates seamlessly with the existing radio frequency front-end⁶.

Strategic Implications and Conclusion

The problem statement SIH26055 issued by the Defence Research and Development Organisation underscores a critical paradigm shift in modern conflict: the necessary transition from static, human-programmed reconnaissance to dynamic, machine-intelligence-driven spectrum dominance. The mathematical and architectural analysis demonstrates definitively that against contemporary threats—including Multi-Function Radars, frequency-agile systems, and Low Probability of Intercept emitters—traditional open-loop scanning guarantees mission failure due to an inherently low, uncontrollable Probability of Intercept.

The proposed software framework directly and exhaustively answers this challenge. By fusing Quantum-Inspired Multi-Armed Bandits for rapid, zero-knowledge spectrum exploration, Predictive State Representations for mapping complex radar behaviors without prior databases, and Dueling Double Deep Q-Networks for optimal, closed-loop temporal resource allocation, the architecture represents the state-of-the-art in cognitive electronic warfare. By continuously minimizing the average intercept time error and maximizing the interception ratio, this artificial intelligence-driven scheduler ensures that tactical receivers maintain an unblinking, predictive awareness of the electromagnetic battlespace, transforming raw intercept data into actionable, mission-critical electronic intelligence.

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